David B. Lindell

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Current Appointments

Assistant Professor
Dept. of Computer Science, University of Toronto — *2022–present*

Faculty Affiliate
Vector Institute — *2022–present*

Faculty Fellow
AXL — *2025–present*

Advisor
Rhoda AI — *2025–present*

Education

Stanford University — **Stanford, CA**
Ph.D. Electrical Engineering — *2016–2021*
Committee: Gordon Wetzstein, Bernd Girod, Mark Horowitz, Vivek Goyal, James Harris

Brigham Young University — **Provo, UT**
M.Sc. Electrical Engineering — *2015–2016*
Advisor: David G. Long

Brigham Young University — **Provo, UT**
B.Sc. Electrical Engineering — *2009–2015*
Advisors: David G. Long, Aaron Hawkins

Previous Professional Experience

LG Electronics Toronto AI Lab — **Toronto, ON**
Principal AI Research Scientist — *2024–2025*

Stanford University — **Stanford, CA**
Postdoctoral Scholar — *2021–2022*
Advisor: Gordon Wetzstein

Intelligent Systems Lab, Intel Corporation — **Santa Clara, CA**
Intern — *2018*
Advisor: Vladlen Koltun

Rincon Research Corporation — **Tucson, AZ**
Intern — *2016*

Awards

CVPR Outstanding Area Chair	2026
CVPR Best Student Paper Award	2025
Ontario Early Researcher Award	2025
Google Research Scholar Award	2024
Sony Focused Research Award	2023
Sony Faculty Innovation Award	2023
David Marr Prize (ICCV Best Paper Award)	2023
Connaught New Researcher Award	2023
ACM SIGGRAPH Outstanding Doctoral Dissertation Honorable Mention	2021
ACM SIGGRAPH Thesis Fast Forward Honorable Mention	2020

CVPR Outstanding Reviewer	2020
Stanford Graduate Research Fellowship	2016–2020
BYU Office of Research & Creative Activities Grant	2015
Tau Beta Pi Scholarship	2014
BYU Heritage Scholarship	2012–2015

Conference Organization/Editorial Positions

Area Chair: Neural Information Processing Systems (NeurIPS)	2023–2025
Area Chair: IEEE Conference on Computer Vision and Pattern Recognition (CVPR)	2023–2026
Associate Editor: ACM Transactions on Graphics	2025–
Associate Editor: IEEE Transactions on Computational Imaging	2023–
Organizer: GeoFreeNVS: Geometry-Free Novel View Synthesis and Controllable Video Models (CVPR)	2026
Organizer: GeoFreeNVS: Geometry-Free Novel View Synthesis and Controllable Video Models (ICCV)	2025
Computational Imaging Technical Committee: IEEE Signal Processing Society	2025–
Finance Co-Chair: Int. Conference on Computational Photography (ICCP)	2022
Organizer: 2nd Workshop on Neural Fields Beyond Conventional Cameras (CVPR)	2025
Program Co-Chair: Int. Conference on Computational Photography (ICCP)	2025
Program Co-Chair: IEEE Workshop on Computational Cameras and Displays (CCD)	2020,2021,2023
Program Committee: Int. Conference on Computational Photography (ICCP)	2019–2024
Technical Papers Committee: SIGGRAPH Asia	2023
Technical Papers Committee: SIGGRAPH	2025

Referee Service

CVPR	2020–
ECCV	2020–
ICCV	2021–
ICCP	2019–
ICLR	2021–
NeurIPS	2021–
SIGGRAPH	2020–
SIGGRAPH Asia	2022–
WACV	2024–
Nature	
Nature Communications	
Nature Photonics	
Optica	
Optics Express	
Science Advances	
IEEE Transactions on Computational Imaging	
IEEE Transactions on Pattern Analysis and Machine Intelligence	

University Service

AI Curriculum Committee: Computer Science Department, University of Toronto	2023–2024
Outreach Committee: Computer Science Department, University of Toronto	2023–2025

Undergraduate Affairs Committee: Computer Science Department, University of Toronto 2022–2024
Faculty Search Committee: Computer Science Department, University of Toronto 2024–2025
Faculty Council Computer Science Department, University of Toronto 2025–2026

Teaching

University of Toronto **Instructor**
CSC2539: Physics-Informed Neural Representations for Visual Computing *2025*

University of Toronto **Instructor**
CSC2529: Computational Imaging *2022–2025*

University of Toronto **Instructor**
CSC420: Introduction to Image Understanding *2023–2025*

AAAI Conference on Artificial Intelligence **Instructor**
AI for Emerging Inverse Problems in Computational Imaging *2024*

Stanford University **Instructor**
EE367/CS448i: Computational Imaging *2022*

Stanford University **Teaching Assistant**
EE367/CS448i: Computational Imaging *2020*

ACM SIGGRAPH **Instructor**
Computational Time-Resolved Imaging, Single-Photon Sensing and Non-Line-of-Sight Imaging *2020*

Journal Publications

[J17] D. Du, A. Xie, P. Mirdehghan, B. Buscaino, S.-H. Baek, K. N. Kutulakos, **D. B. Lindell**, “Polarimetric full-wavefield coherent lidar,” *Optica*, 2026.

[J16] E. Y. Lin, Z. Wang, R. Lin, D. Miau, F. Kainz, J. Chen, X. C. Zhang, **D. B. Lindell**, K. N. Kutulakos, “Learning lens blur fields,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 9, pp. 7383–7395, 2025.

[J15] S. Tedla, K. Zhu, T. Canham, F. Taubner, M. S. Brown, K. N. Kutulakos, **D. B. Lindell**, “Generating the past, present and future from a motion-blurred image,” *ACM Transactions on Graphics* (*TOG*), vol. 44, no. 6, pp. 1–15, 2025.

[J14] D. Verma, I. Ruffolo, **D. B. Lindell**, K. N. Kutulakos, A. Mariakakis, “Chromaflash: Snapshot hyperspectral imaging using rolling shutter cameras,” *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (*Ubicomp*), vol. 8, no. 3, pp. 1–31, 2024.

[J13] C. Shentu, E. Li, C. Chen, P. T. Dewi, **D. B. Lindell**, J. Burgner-Kahrs, “MoSS: Monocular shape sensing for continuum robots,” *IEEE Robotics and Automation Letters*, 2023.

[J12] J. N. P. Martel, **D. B. Lindell**, C. Z. Lin, E. R. Chan, M. Monteiro, G. Wetzstein, “ACORN: Adaptive coordinate networks for neural scene representation,” *ACM Transactions on Graphics* (*SIGGRAPH*), vol. 40, no. 4, pp. 1–13, 2021.

[J11] **D. B. Lindell** and G. Wetzstein, “Three-dimensional imaging through scattering media based on confocal diffuse tomography,” *Nature Communications*, vol. 11, no. 4517, 2020.

[J10] C. A. Metzler, **D. B. Lindell**, G. Wetzstein, “Keyhole imaging: Non-line-of-sight imaging and tracking of moving objects along a single optical path at long standoff distances,” *IEEE Transactions on Computational Imaging*, vol. 7, pp. 1–12, 2020.

[J9] Z. Sun, **D. B. Lindell**, O. Solgaard, G. Wetzstein, “SPADnet: Deep RGB-SPAD sensor fusion assisted by monocular depth estimation,” *Optics Express*, vol. 28, no. 10, pp. 14 948–14 962, 2020.

[J8] F. Heide, M. O’Toole, K. Zang, **D. B. Lindell**, S. Diamond, G. Wetzstein, “Non-line-of-sight imaging with partial occluders and surface normals,” *ACM Transactions on Graphics* (*ToG*), vol. 38, no. 3, 2019.

[J7] **D. B. Lindell**, G. Wetzstein, M. O’Toole, “Wave-based non-line-of-sight imaging using fast f-k migration,” *ACM Transactions on Graphics* (*SIGGRAPH*), vol. 38, no. 4, 2019.

[J6] F. Heide, S. Diamond, **D. B. Lindell**, G. Wetzstein, “Sub-picosecond photon-efficient 3D imaging using single-photon sensors,” *Scientific Reports*, vol. 8, no. 17726, 2018.

[J5] **D. B. Lindell**, M. O’Toole, G. Wetzstein, “Single-photon 3D imaging with deep sensor fusion,” *ACM Transactions on Graphics* (*SIGGRAPH*), vol. 37, no. 4, 2018.

[J4] M. O’Toole, **D. B. Lindell**, G. Wetzstein, “Confocal non-line-of-sight imaging based on the light-cone transform,” *Nature*, vol. 555, no. 7696, pp. 338–341, 2018.

[J3] **D. B. Lindell** and D. G. Long, “High-resolution soil moisture retrieval with ASCAT,” *IEEE Geoscience and Remote Sensing Letters*, vol. 13, no. 7, pp. 972–976, 2016.

[J2] **D. B. Lindell** and D. G. Long, “Multiyear Arctic ice classification using ASCAT and SSMIS,” *Remote Sensing*, vol. 8, no. 4, p. 294, 2016.

[J1] **D. B. Lindell** and D. G. Long, “Multiyear Arctic sea ice classification using OSCAT and QuikSCAT,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 54, no. 1, pp. 167–175, 2016.

Conference Publications

[C47] N. Artru, R. Hussain, E. Got, A. Messier, **D. B. Lindell**, A. Dib, “Skullptor: High fidelity 3d head reconstruction in seconds with multi-view normal prediction,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2026.

[C46] S. Bahmani, T. Shen, J. Ren, J. Huang, Y. Jiang, H. Turki, A. Tagliasacchi, **D. B. Lindell**, Z. Gojcic, S. Fidler, “Lyra: Generative 3D scene reconstruction via video diffusion model self-distillation,” in *International Conference on Learning Representations* (*ICLR*), 2026.

[C45] A. Y. Guo, A. Malik, S. Tedla, Y. Dai, Y. Qin, Z. Salehe, B. Attal, S. Nousias, K. Kutulakos, **D. B. Lindell**, “Dark3r: Learning structure from motion in the dark,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2026.

[C44] A. Malik, D. Chan, X. Zhao, **D. B. Lindell**, O. Tuzel, J.-H. R. C. Chang, “Velox: Learning representations of 4d geometry and appearance,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2026.

[C43] V. Polizzi, S. Yang, Q. Clark, J. Kelly, I. Gilitschenski, **D. B. Lindell**, “VibES: Induced vibration for persistent event-based sensing,” in *International Conference on 3D Vision* (*3DV*), 2026.

[C42] H. Qiu, K. N. Kutulakos, **D. B. Lindell**, “Efficient and training-free single-image diffusion models,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2026.

[C41] D. Verma, A. Qiu, R. Rangel, A. Barman, H. Yang, C. Hu, F. Zhang, R. Genov, **D. B. Lindell**, K. N. Kutulakos, “Lumosaic: Hyperspectral video via active illumination and coded-exposure pixels,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2026.

[C40] A. Xie, D. Du, S. Nousias, **D. B. Lindell**, K. Kutulakos, “Inter-photon-limited videography,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2026.

[C39] M. Anagh, B. Attal, A. Xie, M. O’Toole, **D. B. Lindell**, “Neural inverse rendering from propagating light,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2025, **(Best Student Paper — 1 of 2 awards, 13,008 submissions)**.

[C38] S. Bahmani, I. Skorokhodov, G. Qian, A. Siarohin, W. Menapace, A. Tagliasacchi, **D. B. Lindell**, S. Tulyakov, “AC3D: Analyzing and improving 3D camera control in video diffusion transformers,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2025.

[C37] S. Bahmani, I. Skorokhodov, A. Siarohin, W. Menapace, G. Qian, M. Vasilkovsky, H.-Y. Lee, C. Wang, J. Zou, A. Tagliasacchi, **D. B. Lindell**, S. Tulyakov, “VD3D: Taming large video diffusion transformers for 3D camera control,” in *International Conference on Learning Representations* (*ICLR*), 2025.

[C36] K. Namekata, S. Bahmani, Z. Wu, Y. Kant, I. Gilitschenski, **D. B. Lindell**, “SG-I2V: Self-guided trajectory control in image-to-video generation,” in *International Conference on Learning Representations* (*ICLR*), 2025.

[C35] S. Nousias, M. Wei, H. Xiao, M. Wu, S. Athar, K. J. Wang, A. Malik, D. A. Barmherzig, **D. B. Lindell**, K. Kutulakos, “Opportunistic single-photon time of flight,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2025, **(Oral Presentation — Top 0.7%, 13,008 submissions)**.

[C34] S. Shekarforoush, **D. B. Lindell**, M. A. Brubaker, D. J. Fleet, "Reconstructing heterogeneous biomolecules via hierarchical gaussian mixtures and part discovery," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[C33] F. Taubner, R. Zhang, M. Tuli, S. Bahmani, **D. B. Lindell**, "MVPD4D: Multi-view portrait video diffusion for animatable 4D avatars," in *Proceedings of SIGGRAPH Asia*, 2025.

[C32] F. Taubner, R. Zhang, M. Tuli, **D. B. Lindell**, "CAP4D: Creating animatable 4D portrait avatars with morphable multi-view diffusion models," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, **(Oral Presentation — Top 0.7%, 13,008 submissions)**.

[C31] V. Zehtab, **D. B. Lindell**, M. A. Brubaker, M. S. Brown, "Efficient neural network encoding for 3d color lookup tables," in *AAAI Conference on Artificial Intelligence*, 2025.

[C30] S. Bahmani, X. Liu, Y. Wang, I. Skorokhodov, V. Rong, Z. Liu, X. Liu, J. J. Park, S. Tulyakov, G. Wetzstein, A. Tagliasacchi, **D. B. Lindell**, "TC4D: Trajectory-conditioned text-to-4D generation," in *European Conference on Computer Vision (ECCV)*, 2024.

[C29] S. Bahmani, I. Skorokhodov, V. Rong, G. Wetzstein, L. Guibas, P. Wonka, S. Tulyakov, J. J. Park, A. Tagliasacchi, **D. B. Lindell**, "4D-fy: Text-to-4D generation using hybrid score distillation sampling," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[C28] W. Luo, A. Malik, **D. B. Lindell**, "Transientangelo: Few-viewpoint surface reconstruction using single-photon lidar," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.

[C27] A. Malik, N. Juravsky, R. Po, G. Wetzstein, K. N. Kutulakos, **D. B. Lindell**, "Flying with photons: Rendering novel views of propagating light," in *European Conference on Computer Vision (ECCV)*, 2024, **(Oral Presentation — Top 2.3%, 8585 submissions)**.

[C26] P. Mirdehghan, B. Buscaino, M. Wu, D. Charlton, M. E. Mousa-Pasandi, K. N. Kutulakos, **D. B. Lindell**, "Coherent optical modems for full-wavefield lidar," in *ACM SIGGRAPH Asia*, 2024.

[C25] P. Mirdehghan, M. Wu, W. Chen, **D. B. Lindell**, K. N. Kutulakos, "TurboSL: Dense, accurate and fast 3D by neural inverse structured light," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[C24] R. Rangel, X. Sun, A. Barman, R. Gulve, S. Bajic, J. Wang, H. Wang, **D. B. Lindell**, K. N. Kutulakos, R. Genov, "23,000-exposures/s 360fps-readout software-defined image sensor with motion-adaptive spatially varying imaging speed," in *IEEE Symposium on VLSI Technology and Circuits*, 2024.

[C23] V. Rong, J. Chen, S. Bahmani, K. N. Kutulakos, **D. B. Lindell**, "GStex: Per-primitive texturing of 2D Gaussian splatting for decoupled appearance and geometry modeling," in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.

[C22] S. Shekarforoush, **D. B. Lindell**, M. A. Brubaker, D. J. Fleet, "CryoSPIN: Improving ab-initio cryo-EM reconstruction with semi-amortized pose inference," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[C21] K. Yin, V. Rao, R. Jiang, X. Liu, P. Aarabi, **D. B. Lindell**, "SCE-MAE: Selective correspondence enhancement with masked autoencoder for self-supervised landmark estimation," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[C20] R. Gulve, R. Rangel, A. Barman, D. Nguyen, M. Wei, M. A. Sakr, X. Sun, **D. B. Lindell**, K. N. Kutulakos, R. Genov, "Dual-port CMOS image sensor with regression-based HDR flux-to-digital conversion and 80 ns rapid-update pixel-wise exposure coding," in *IEEE International Solid-State Circuits Conference (ISSCC)*, 2023.

[C19] A. Malik, P. Mirdehghan, S. Nousias, K. N. Kutulakos, **D. B. Lindell**, "Transient neural radiance fields for lidar view synthesis and 3d reconstruction," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023, **(Spotlight — Top 3.6%, 12345 submissions)**.

[C18] S. Sinha, J. Y. Zhang, A. Tagliasacchi, I. Gilitschenski, **D. B. Lindell**, "SparsePose: Sparse-view camera pose regression and refinement," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[C17] M. Wei, S. Nousias, R. Gulve, **D. B. Lindell**, K. N. Kutulakos, "Passive ultra-wideband single-photon imaging," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, **(Marr Prize — awarded best paper out of 8068 submissions)**.

[C16] A. W. Bergman, P. Kellnhofer, Y. Wang, E. R. Chan, **D. B. Lindell**, G. Wetzstein, "Generative neural articulated radiance fields," in *Advances in Neural Information Processing Systems* (*NeurIPS*), 2022.

[C15] C. Z. Lin, **D. B. Lindell**, E. R. Chan, G. Wetzstein, "3D GAN inversion for controllable portrait image animation," in *ECCV 2022 Workshop on Learning to Generate 3D Shapes and Scenes*, 2022.

[C14] **D. B. Lindell**, D. Van Veen, J. J. Park, G. Wetzstein, "BACON: Band-limited coordinate networks for neural scene representation," in *IEEE Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2022, **(Oral Presentation — Top 4.1%, 8262 submissions)**.

[C13] S. Shekarforoush, **D. B. Lindell**, D. J. Fleet, M. A. Brubaker, "Residual multiplicative filter networks for multiscale reconstruction," in *Advances in Neural Information Processing Systems* (*NeurIPS*), 2022.

[C12] D. Van Veen, R. Van der Sluijs, B. Ozturkler, A. D. Desai, C. Bluethgen, R. D. Boutin, M. H. Willis, G. Wetzstein, **D. B. Lindell**, S. Vasanawala, J. Pauly, A. S. Chaudhari, "Scale-agnostic super-resolution in MRI using feature-based coordinate networks," in *Medical Imaging with Deep Learning*, 2022.

[C11] Q. Zhao, **D. B. Lindell**, G. Wetzstein, "Learning to solve PDE-constrained inverse problems with graph networks," in *International Conference on Machine Learning* (*ICML*), 2022.

[C10] **D. B. Lindell**, J. N. P. Martel, G. Wetzstein, "AutoInt: Automatic integration for fast neural volume rendering," in *IEEE Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2021.

[C9] A. W. Bergman, **D. B. Lindell**, G. Wetzstein, "Deep adaptive LiDAR: End-to-end optimization of sampling and depth completion at low sampling rates," in *IEEE International Conference on Computational Photography* (*ICCP*), 2020.

[C8] **D. B. Lindell**, M. O'Toole, G. Wetzstein, "Efficient non-line-of-sight imaging with computational single-photon imaging," in *Advanced Photon Counting Techniques XIV*, SPIE, 2020.

[C7] **D. B. Lindell** and G. Wetzstein, "Confocal diffuse tomography for single-photon 3D imaging through highly scattering media," in *Computational Optical Sensing and Imaging* (*COSI*), OSA, 2020.

[C6] M. Nishimura, **D. B. Lindell**, C. A. Metzler, G. Wetzstein, "Disambiguating monocular depth estimation with a single transient," in *European Conference on Computer Vision* (*ECCV*), 2020.

[C5] V. Sitzmann, J. N. P. Martel, A. W. Bergman, **D. B. Lindell**, G. Wetzstein, "Implicit neural representations with periodic activation functions," in *Advances in Neural Information Processing Systems* (*NeurIPS*), 2020, **(Oral Presentation — Top 1.1%, 9467 submissions)**.

[C4] S. I. Young, **D. B. Lindell**, B. Girod, D. Taubman, G. Wetzstein, "Non-line-of-sight surface reconstruction using the directional light-cone transform," in *IEEE Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2020, **(Oral Presentation — Top 5.2%, 6424 submissions)**.

[C3] **D. B. Lindell**, G. Wetzstein, V. Koltun, "Acoustic non-line-of-sight imaging," in *IEEE Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2019, **(Oral Presentation — Top 5.5%, 5165 submissions)**.

[C2] **D. B. Lindell**, M. O'Toole, G. Wetzstein, "Towards transient imaging at interactive rates with single-photon detectors," in *IEEE International Conference on Computational Photography* (*ICCP*), 2018.

[C1] M. O'Toole, F. Heide, **D. B. Lindell**, K. Zang, S. Diamond, G. Wetzstein, "Reconstructing transient images from single-photon sensors," in *IEEE Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2017, **(Spotlight — Top 8.0%, 2680 submissions)**.

Public Demonstrations

[D3] R. Gulve, R. Rangel, A. Barman, D. Nguyen, M. Wei, M. A. Sakr, X. Sun, **D. B. Lindell**, K. N. Kutulakos, R. Genov, "Low-cost coded-exposure pixel cameras for robust high-speed computational imaging at up to 18,000 exposures per second," in *CVPR Demos*, 2023.

[D2] M. O'Toole, **D. B. Lindell**, G. Wetzstein, "Real-time non-line-of-sight imaging," in *ACM SIGGRAPH Emerging Technologies*, 2018.

[D1] M. O'Toole, **D. B. Lindell**, G. Wetzstein, "Real-time non-line-of-sight imaging," in *CVPR Demos*, 2018.

Theses

2021: Computational Imaging with Single-Photon Detectors. Ph.D. Thesis.

2016: Arctic Sea Ice Classification and Soil Moisture Estimation Using Microwave Sensors. Master's Thesis.

Keynotes

2025: Inverse Rendering from Propagating Light, GraphQuon 2025, Toronto, ON

2025: Inverse Rendering from Propagating Light, IEEE Conference on Computational Imaging Using Synthetic Apertures, College Park, Maryland

2024: Flying with Photons: Rendering Novel Views of Propagating Light, 1st Workshop on Neural Fields Beyond Conventional Cameras (ECCV 2024), Milan, Italy

Invited Talks

2026: Foundation Models for Computational Photography, Sony America Seminar Series, San Jose, CA (Virtual)

2025: Inverse Rendering from Propagating Light, SFU Intelligent Visual Computing and Consumer Device Summit, Burnaby, BC (Virtual)

2025: Inverse Rendering from Propagating Light, ICCV Workshop on Computer Vision with Single-Photon Cameras, Honolulu, HI

2025: Inverse Rendering from Propagating Light, ICCV Unilight Workshop, Honolulu, HI

2025: Inverse Rendering from Propagating Light, Waabi, Toronto, ON

2025: Acquisition-Aware Generative Scene Reconstruction, Great Lakes Graphics Workshop, Chicago, IL

2025: Inverse Rendering from Propagating Light, SIGGRAPH TPC Workshop, Vancouver, BC

2025: From Pixels to Perception: Artificial Intelligence and Computer Vision, CS Academy, Toronto, ON

2025: Capturing Dynamic Scenes from Seconds to Picoseconds, MERL, Virtual

2024: Capturing Dynamic Scenes from Seconds to Picoseconds, Computer Vision Seminar, Seoul National University, Seoul, South Korea

2024: Capturing Dynamic Scenes from Seconds to Picoseconds, Graphics Seminar, POSTECH University, Pohang, South Korea

2024: Neural Scene Reconstruction from Videos of Propagating Light, Korea AI Summit, Seoul, South Korea

2024: Capturing Dynamic Scenes from Seconds to Picoseconds, Computational Imaging Seminar, Sony Corporation, Tokyo, Japan

2024: Capturing Dynamic Scenes from Seconds to Picoseconds, Computer Vision Seminar, University of Tokyo, Tokyo, Japan

2024: Imaging Anytime Anywhere All at Once: Capturing Dynamic Scenes from Seconds to Picoseconds, UC Berkeley Photobears, Berkeley, CA

2024: Imaging Anytime Anywhere All at Once: Capturing Dynamic Scenes from Seconds to Picoseconds, Stanford Center for Image Systems Engineering (SCIEN), Stanford, CA

2024: Flying with Photons: Rendering Novel Views of Propagating Light, Conference on Robots and Vision, Guelph, ON

2024: Text-to-4D Generation Using Hybrid Score Distillation Sampling, UTMIST Immersion Night, Toronto, ON

2024: Passive Ultra-Wideband Single-Photon Imaging, Stanford EE367 Computational Imaging (Guest Lecture), Virtual

2024: From Pixels to Perception: Artificial Intelligence and Computer Vision, DGP Academy, Toronto, ON

2024: Passive Ultra-Wideband Single-Photon Imaging, Simon Fraser University (GrUVi Lab), Vancouver, BC

2024: Passive Ultra-Wideband Single-Photon Imaging, National Research Council Ultrafast Quantum Photonics Group, Ottawa, ON

2023: Passive Ultra-Wideband Single-Photon Imaging, 3rd International Computational Imaging Conference, Virtual

2023: From Pixels to Perception: Artificial Intelligence and Computer Vision, Leadership by Design Workshop, Toronto, ON

2023: Neural Rendering at One Trillion Frames per Second, UTMIST EigenAI Conference, Toronto, ON

2023: Passive Ultra-Wideband Single-Photon Imaging, Photons Canada, Virtual

2023: Neural Rendering at One Trillion Frames per Second, BIRS Workshop on Generative 3D Models, Banff, AB

2023: Passive Ultra-Wideband Single-Photon Imaging, Photonics North, Montreal, QC

2023: Physics-Based Visual Computing for Efficient 3D Vision and Sensing, University of Windsor, Windsor, ON

2022: Recent Advances in Non-Line-of-Sight Imaging, IEEE Signal Processing Society Webinar, Virtual

2022: Physics-Based Visual Computing for Efficient 3D Vision and Sensing, Purdue Computational Imaging Seminar, Virtual

2022: Confocal Non-Line-of-Sight Imaging and Diffuse Tomography Using Single-Photon Sensors, Imaging and Applied Optics Congress, Vancouver, BC

2022: Physics-Based Visual Computing for Efficient 3D Vision and Sensing, Caltech, Pasadena, CA

2022: Implicit Neural Representation Networks for Fitting Signals, Derivatives, and Integrals, Silicon Valley ACM SIGGRAPH Chapter, Virtual

2021: Implicit Neural Representation Networks for Fitting Signals, Derivatives, and Integrals, Samsung AI Centre, Virtual

2021: Implicit Neural Representation Networks for Fitting Signals, Derivatives, and Integrals, University of Erlangen-Nuremberg, Virtual

2021: Physics-Based Visual Computing for Efficient 3D Vision and Sensing, University of Michigan, Virtual

2021: Physics-Based Visual Computing for Efficient 3D Vision and Sensing, MIT RLE, Virtual

2021: Physics-Based Visual Computing for Efficient 3D Vision and Sensing, University of Chicago, Virtual

2021: Physics-Based Visual Computing for Efficient 3D Vision and Sensing, University of Toronto, Virtual

2021: Physics-Based Visual Computing for Efficient 3D Vision and Sensing, Texas A&M, Virtual

2021: AutoInt: Automatic Integration for Fast Neural Volume Rendering, Google, Virtual

2020: Implicit Neural Representation Networks for Fitting Signals, Derivatives, and Integrals, Graphics and Mixed Environment Seminar (GAMES), Virtual

2020: A Camera to See Around Corners, Playground/Akasha Imaging, Palo Alto, CA

2019: A Camera to See Around Corners, TEDxBeaconStreet, Boston, MA

2019: Computational Imaging with Single-Photon Detectors, Boston University Center for Information & Systems Engineering (CISE), Boston, MA

2019: Efficient Confocal Non-Line-of-Sight Imaging, MIT RLE, Cambridge, MA

2019: Efficient Confocal Non-Line-of-Sight Imaging, MIT Media Lab, Cambridge, MA

2019: Computational Imaging with Single-Photon Detectors, Berkeley Center for Computational Imaging, Berkeley, CA

2019: Computational Single-Photon Imaging, Silicon Valley ACM SIGGRAPH Chapter, San Jose, CA

2019: Computational Imaging with Single-Photon Detectors, Stanford Center for Image Systems Engineering (SCIEN), Stanford, CA

2019: Computational Single-Photon Imaging, Carnegie Mellon University Graphics Lab, Pittsburgh, PA

Press Coverage and Appearances

Podcasts

2025: Computational Imaging. *Carry the Two*, IMSI

2022: BACON: Band-Limited Coordinate Networks. *Talking Papers Podcast*

Selected Press

2024: AI algorithm captures photons in motion. *Phys.org*

2024: Flying with Photons: Rendering Novel Views of Propagating Light. *Veritasium* (YouTube)

2024: Flying with Photons. *University of Toronto News*

2023: U of T computer scientists develop video camera that acts as a microscope for time. *University of Toronto News*

2023: Single-photon camera enables extreme data acquisition for computer vision. *MathWorks*

2020: Seeing objects through clouds and fog. *Stanford News*

2020: Scientists have created a laser scanner that can see through fog. *Gizmodo*

2019: Scientists use sound to see around corners. *Science Magazine*

2019: New camera can film moving object from around a corner. *Stanford News*

2019: Looking around corners with f–k migration. *Seeker* (YouTube)

2018: The new science of seeing around corners. *Quanta Magazine*

2018: Wanna see around corners? Better get yourself a laser. *Wired*

2018: Self-driving cars may soon be able to see around corners. *The Guardian*

2018: New technology can see round corners like magic. *Newsweek*

2018: Autonomous cars could peep around corners via bouncing laser. *TechCrunch*

2018: Driverless cars may soon see round corners using super laser. *The Telegraph*

2018: Confocal non-line-of-sight imaging based on the light-cone transform. *TED.com*

Postdoctoral Advising

Dongyu Du — March 2025–
University of Toronto

Ben Attal — Sep 2025–
University of Toronto

PhD Advising

Haojun Qiu — Sep 2026–
University of Toronto

Ruihang Zhang — Sep 2026–
University of Toronto

Sophia Luo — Sep 2026–
University of Toronto

Willy (Ding-Jiun) Huang — Sep 2026–
University of Toronto

Victor Chu — Sep 2025–
University of Toronto

Michael Neumayr — Sep 2025–
University of Toronto

Sophia Yang — Sep 2025–
University of Toronto

Andrew Guo — Sep 2024–
University of Toronto

Felix Taubner — Sep 2024–
University of Toronto

Sherwin Bahmani — Sep 2023–April 2026
University of Toronto

Victor Rong — Sep 2023–
University of Toronto
Co-advised with Kyros Kutulakos

Anagh Malik — Sep 2022–
University of Toronto

Samarth Sinha — Sep 2022–
University of Toronto
Co-advised with Igor Gilitschenski

Esther Lin — Sep 2022–
University of Toronto
Co-advised with Kyros Kutulakos

Master's Advising

Abhinav Shankar H — May 2026–
MScAC, University of Toronto

Noé Artru — May 2025–Dec 2025
MScAC, University of Toronto
Ubisoft

Andrew Xie — Sep 2024–May 2026
MSc, University of Toronto

Kelly Zhu — Sep 2024–May 2026
MSc, University of Toronto

Maxx Wu — Sep 2023–May 2025
MSc, University of Toronto
Co-advised with Kyros Kutulakos

Umar Masud — May 2024–Dec 2024
MScAC, University of Toronto
Samsung

Steven Hyun — May 2024–Dec 2024
MScAC, University of Toronto
Samsung

Faraz Ali — May 2024–Dec 2024
MScAC, University of Toronto
Samsung

Carolina Villamizar — May 2024–Dec 2024
MScAC, University of Toronto
DWave

Kartik Kumar — May 2023–Dec 2023
MScAC, University of Toronto
DNEG

Yihan (Nick) Ni — May 2023–Dec 2023
MScAC, University of Toronto
DNEG

Kejia Yin — May 2023–Dec 2023
MScAC, University of Toronto
MODIFACE

Vahid Zehtab — May 2023–Dec 2023
MScAC, University of Toronto

Samsung

EJay Guo May 2022–Dec 2022
MScAC, University of Toronto
DNEG

Visiting Students

Felix Lange Sep 2026–May 2027
Free University of Berlin

Ugo Cavalcanti June 2026–Nov 2026
University of Bologna

Jorge Garcia Pueyo March 2026–June 2026
University of Zaragoza
Co-advised with Adolfo Muñoz

Hoon-Gyu Chung Sep 2025–June 2026
POSTECH

Seung-Hwan Baek June 2025–Aug 2025
POSTECH

Michael Neumayr Sep 2024–Apr 2025
TUM
Co-advised with Matthias Nießner

Siddharth Somasundaram Sep 2024–Dec 2024
MIT
Co-advised with Kyros Kutulakos, Ramesh Raskar

Undergraduate Advising

Pratham Gupta May 2026–Aug 2026
University of Toronto

Prashanth Shyamala Jan 2026–
University of Toronto

Sicheng (Harry) Zhou Sep 2025–
University of Toronto

Junhao (Harry) Wang Sep 2025–May 2026
University of Toronto

Ruihang Zhang Sep 2024–May 2026
University of Toronto

Allison Lau Sep 2024–Dec 2024
University of Toronto

Koichi Namekata July 2024–Dec 2024
University of Toronto

Howard Xiao May 2024–Aug 2025
University of Toronto
NSERC Undergraduate Student Research Award

Mehar Khurana May 2024–Aug 2024
IIT Madras
Mitacs Globalink

Weihan Luo Sep 2023–Aug 2025
University of Toronto

Zixin Guo Jan 2023–Dec 2024
University of Toronto

University of Toronto Excellence Award

Zach Salehe Jan 2023–May 2024
University of Toronto
NSERC Undergraduate Student Research Award

Jason Zhu Sep 2023–May 2024
University of Toronto

Steven Luo Jan 2024–May 2024
University of Toronto

Ariel Chen Sep 2023–May 2024
University of Toronto

Haojun Qiu Sep 2022–May 2024
University of Toronto

Andrew Qiu Sep 2023–Dec 2023
University of Toronto

Kevin Vaidyan May 2023–May 2024
University of Toronto

Noah Juravsky May 2023–May 2024
University of Toronto

Dorsa Molaverdikhani Jan 2023–May 2024
University of Toronto

Shahmeer Athar Jan 2023–May 2024
University of Toronto

Rishit Dagli Jan 2023–Dec 2023
University of Toronto

Roland Gao Jan 2023–Sep 2023
University of Toronto
NSERC Undergraduate Student Research Award

Qing (Amy) Lyu Jan 2023–May 2023
University of Toronto

Louis Zhang Jan 2023–May 2023
University of Toronto

Junru Lin Sep 2022–May 2023
University of Toronto

Justin Tran Sep 2022–May 2023
University of Toronto
Thesis: Generative 3D shape modeling using latent space diffusion

Skyler Zhang Sep 2022–May 2023
University of Toronto
Thesis: Towards coded high-speed video acquisition using diffusion models

Funding

Google Research Gift Dec 2025 –
Inverse Rendering from Propagating Light with Generative Priors

NSERC Alliance Sep 2025 – Sep 2026
Computational Imaging using Coded-Exposure Sensors and Flux-to-Digital Conversion

NSERC Alliance Jul 2025 – Jul 2026
Multiview Burst Imaging for High Dynamic Range and Low-Light Radiance Fields

Ontario Early Researcher Award — Jun 2025 – Jun 2031
Single-Photon Imaging with Large Generative Models

NSERC Alliance — Dec 2024 – Dec 2027
Generative Video Models for Controllable Digital Human Avatars

NSERC Alliance–Mitacs Grant — Jul 2024 – July 2028
Implicit Representations for 4D Digital Humans

Sony Focused Research Award — Sep 2024 – Sep 2025
Multiview Burst Imaging for High Dynamic Range and Low-Light Neural Radiance Fields

Sony Faculty Innovation Award — Sep 2024 – Sep 2025
Differentiable Computational Imaging Using Coded Exposure Sensors and Flux-to-Digital Conversion

LG Research Grant — May 2024 – April 2025
Learning Controllable 4D Avatars from Portrait Video Collections

Google Research Scholar Award — Apr 2024 –
Computational Single-Photon Photography for Dynamic Vision in the Dark

Snap Inc. Research Gift — Feb 2024 –
Camera Control in 2D Video Generators

NSERC Alliance–Mitacs Grant — Jan 2024 – Jan 2026
Ultrafast Phase-Modulated Coherent Lidar Using Off-the-Shelf Optical Modems

CFI Infrastructure Operating Fund — Dec 2023 – Dec 2028
Neural Signal Representations for Active 3D Imaging

NSERC Research Tools and Instruments Grant — Sep 2023 – Sep 2024
Single-Photon Cameras for Extreme Computer Vision and Computational Astronomy

XSeed/TRANSFORM HF Grant — Sep 2023 – Sep 2025
Towards Home Monitoring of Heart Failure Patients via Robust and Unbiased Spatial Frequency Domain Imaging

Connaught New Researcher Award — Apr 2023 – Apr 2025
Neural Signal Representations for Active 3D Imaging

NSERC Discovery Launch Supplement — Apr 2022 – Apr 2027
Neural Signal Representations for Physics-Based Machine Learning and Active 3D Imaging

NSERC Discovery Grant — Apr 2022 – Apr 2027
Neural Signal Representations for Physics-Based Machine Learning and Active 3D Imaging

Canada Foundation for Innovation John R. Evans Leaders Fund — Sep 2022 – Mar 2026
Neural Signal Representations for Active 3D Imaging